

GLOBAL RESOURCE SYSTEMS

The Global Resource Systems undergraduate major employs a truly interdisciplinary and systemic approach to understanding complex global resource issues. Students develop a core set of technical competencies in a resource area selected from the majors, minors and certificates offered by Iowa State University. Students choose a world region in which to specialize, develop competency in a relevant world language, and participate in a significant cross-cultural internship experience. They carry out a senior project related to their resource specialization within the context of the world region. The undergraduate experience culminates with a senior capstone course, where students work with real-world clients to address global resource challenges.

Multidisciplinary themes are developed in the context of the physical, biological and socio-economic factors affecting global resource systems. In this context, resource systems include natural, food and agricultural, environmental, cultural and human, political and institutional, financial and built, public health and social resources. Graduates of this program have transnational leadership skills and are successful integrators of various specializations on a team. They are skilled in applying a systemic perspective and developing solutions to complex global resource systems problems using innovativeness and creativity. Future professionals communicate effectively and demonstrate environmental awareness, exhibit an ethical perspective, and display clear analysis of how cultural diversity impacts work both here and abroad.

A degree in Global Resource Systems opens the door to employment opportunities in the many businesses and organizations that require globally competent employees.

Student Learning Outcomes

Upon graduation, students should be able to:

Understand sustainable global resource systems by: summarizing factors of biological, physical, and social resources in global systems; and predicting the consequences of the utilization and distribution of global resources and their systems.

Apply global competency skills by recognizing diverse cultures, reflecting on the value of cultures on global resource systems, and employing skills needed to work in different cultures.

Design ethical and innovative solutions to global challenges by using information literacy skills to define global challenges and creating sustainable solutions to global challenges.

Use communication and leadership skills by communicating effectively with diverse audiences using written, oral, visual, and electronic skills and participating effectively as leaders in teams and organizations.

Curriculum in Global Resource Systems

Administered by a supervisory committee in the College of Agriculture and Life Sciences. Students choose a region of the world to develop an expertise; they choose a language to learn and develop proficiency through the intermediate level; they choose and possess an area of technical expertise by completing an additional major, minor or certificate program offered at Iowa State University; they complete a required internship in an international setting; and they select and complete a senior research project with faculty mentoring.

Total Degree Requirement: 125 cr.

Only 65 cr. from a two-year institution may apply which may include up to 16 technical cr.; 9 P-NP cr. of free electives; 2.00 minimum GPA.

International Perspective: 3 cr.

Met by Global Resource Systems requirements

U.S. Cultures and Communities: 3 cr.

3 cr. from approved list 3

Communications Proficiency:

English composition (6 credits with a grade of C or higher; see courses below.)

Speech fundamentals (3 credits with a grade of C or higher; see courses below.)

Communication/Library: 13 cr.

ENGL 1500	Critical Thinking and Communication	3
ENGL 2500	Written, Oral, Visual, and Electronic Composition	3
LIB 1600	Introduction to College Level Research	1
SPCM 2120	Fundamentals of Public Speaking	3
	or AGEDS 3110 Presentation and Sales Strategies for Agricultural Audiences	
	or COMST 214C Professional Communication	
One of the following advanced communications courses:		3
ENGL 3020	Business Communication	
ENGL 3090	Proposal and Report Writing	
ENGL 3140	Technical Communication	

Total Credits 13

Humanities and Social Sciences: 6 cr.

ECON 1010	Principles of Microeconomics	3
	or ECON 1020 Principles of Macroeconomics	
Plus three credit hours from approved humanities list		3

Total Credits 6

Ethics: 3 cr.

3 cr. from approved list 3

Biological Sciences: 3 cr.

Choose one of the following Biological Sciences:

BIOL 1010	Introductory Biology	3
BIOL 2110	Principles of Biology I	
BIOL 2120	Principles of Biology II	
BIOL 2510	Biological Processes in the Environment	

Life Sciences: 3 cr.

3 cr. from approved list

Mathematics and Statistics: 6 cr.

3 cr. of mathematics from approved list

3 cr. of statistics from approved list	3
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Total Credits	6
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physical sciences: 8 cr.

5 credits of physical science (e.g., chemistry, geological and atmospheric sciences, physics) from college-approved list

One course from the following Earth Sciences:	3
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AGRON 1820	Introduction to Soil Science	
or AGRON 2060	Introduction to Weather and Climate	
or AGRON 2820	Soil Conservation and Land Use	
or GEOL 1010	Environmental Geology: Earth in Crisis	
or GEOL 1600	Water Resources of the World	
or GLOBE 1200	Geography of Global Resource Systems	

Total Credits	8
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Global Resource Systems: 23 cr.

GLOBE 1100	Orientation	1
GLOBE 2010	Introduction to Global Resource Systems	3
GLOBE 2110	Issues in Global Resource Systems (Each offering is 1 cr., must be repeated for 3 cr.)	3
GLOBE 3030	Agricultural, Food and Natural Global Resource Systems	3
GLOBE 3040	Socio-Economic Global Resource Systems	3
GLOBE 3200	Global Resource Systems Internship Preparation	1
GLOBE 3210	Internship - Global	3-6
or GLOBE 3220	Internship - United States	
GLOBE 4010	Senior Project	3
GLOBE 4020	Responses to Global Resource System Challenges	3

Total Credits	23-26
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Global Competency: 15 cr.

15 cr. in global competency courses from the university's list of international perspective courses. Up to 3 cr. may be earned from a faculty-led travel course. Any course taken during a semester-long study abroad program may apply.

WORLD LANGUAGE: 16 cr.

Proficiency in a single world language through the intermediate level (e.g., Spanish 1020, 2010, 2010, 2020). Placement or transfer credit in a higher level may meet this requirement in full.

Specialization: 15 cr. or more

Satisfied by any of the majors, minors or certificates offered at Iowa State University.

Electives:

Sufficient coursework to ensure a total of no less than 125 credits.

Global Resource Systems, B.S.

First Year

Fall	Credits Spring	Credits
GLOBE 1100	1 GLOBE 2010	3
ENGL 1500	3 GLOBE 2110	1
LIB 1600	1 ENGL 2500	3
GLOBE 2110	1 ECON 1010 or 1020	3
GLOBAL COMPETENCY ELECTIVE	3 US CULTURES AND COMMUNITIES ELECTIVE	3
BIOLOGICAL SCIENCE ELECTIVE	3 EARTH SCIENCE ELECTIVE	3
SPECIALIZATION ELECTIVE	3	
	15	16

Second Year

Fall	Credits Spring	Credits
GLOBE 3030	3 GLOBE 3040	3
LANGUAGE 1010	4 LANGUAGE 1020	4
STATISTICS ELECTIVE	3-4 HUMANITIES ELECTIVE	3
GLOBAL COMPETENCY	3 SPECIALIZATION ELECTIVES	6
SPECIALIZATION ELECTIVE	3	
	16-17	16

Third Year

Fall	Credits Spring	Credits
LANGUAGE 2010	4 GLOBE 3200	1
GLOBAL COMPETENCY	3 GLOBE 2110	1
PHYSICAL SCIENCE ELECTIVES	5 LANGUAGE 2020	4
SPECIALIZATION ELECTIVES	6 ETHICS ELECTIVE	3
	SPECIALIZATION ELECTIVES	6
	18	15

Fourth Year

Fall	Credits Spring	Credits
GLOBE 3210	3 GLOBE 4020	3
GLOBE 4010	3 ENGL 3020, 3090, or 3140	3
SPEECH ELECTIVE	3 GLOBAL COMPETENCY	3
GLOBAL COMPETENCY ELECTIVE	3 SPECIALIZATION ELECTIVES	6
SPECIALIZATION ELECTIVE	3	
	15	15

Total Credits: 126-127